5. a) Let us show that

$$f: \mathbb{R} \to \mathbb{R}, \qquad x \longmapsto \sum_{n\geq 1} s_n \operatorname{sign}(x-a_n)$$

is indeed a well-defined function, i.e., that $f(x) \in \mathbb{R}$ for all $x \in \mathbb{R}$.

This follows from the absolute convergence of $\sum_{n\geq 1} s_n$, since for any $x \in \mathbb{R}$,

$$\left|\sum_{n\geq 1} s_n \cdot \operatorname{sign}(x-a_n)\right| \leq \sum_{n\geq 1} |s_n| < +\infty .$$

Let now $\varepsilon > 0$, $x_0 \in \mathbb{R}\setminus A$ and $N \in \mathbb{N}$ s.t. $\sum_{n\geq N+1} |s_n| < \varepsilon$ which is again possible by absolute convergence. Choosing $\delta > 0$ sufficiently small, we have

$$a_n \notin (x_0-\delta, x_0+\delta) \qquad \text{for } n \leq N.$$

($\delta := \min\left\{\frac{|x_0-a_n|}{2} : n \leq N\right\}$ works). Then, for all $y \in \mathbb{R}$ s.t. $|x-y| < \delta$ we have

$$|f(x_0)-f(y)| = \left|\sum_{n\geq 1} s_n \overbrace{\left(\operatorname{sign}(x_0-a_n)-\operatorname{sign}(y-a_n)\right)}^{=0 \text{ for all } n\leq N}\right| \leq \sum_{n\geq N} |s_n| < \varepsilon,$$

so f is continuous in x_0.

Now we investigate points $a_m \in A$. Let us define

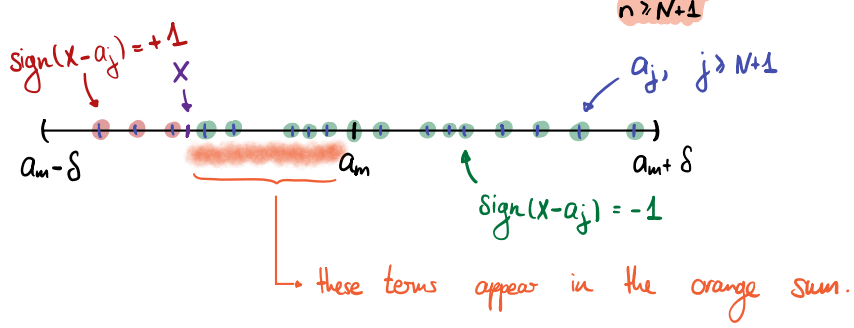
$$A_m^- := \{n \in \mathbb{N} : a_n < a_m\}, \qquad A_m^+ := \{n \in \mathbb{N} : a_n > a_m\}.$$

We claim that $\lim_{x \nearrow a_m} f(x) = \underbrace{\sum_{n\in A_m^-} s_n - \sum_{n\in A_m^+} s_n - s_m}_{=:L}$ and similarly, $\lim_{x\searrow a_m} f(x) = \sum_{n\in A_m^-} s_n - \sum_{n\in A_m^+} s_n + s_m$.

For ε, N and δ as above, we have for $x \in \mathbb{R}$ such that $a_m - x < \delta$,

$$|f(x) - L| = \left|\sum_{n\geq 1} s_n \operatorname{sign}(x-a_n) - L\right| \leq \sum_{\substack{n\in A_m^-,\\ n\geq N+1}} |s_n| \leq \sum_{n\geq N+1} |s_n| < \varepsilon$$

Picture:



The proof for $\lim_{x\searrow a_m} f(x)$ is similar. Putting both one-sided limits together, we get

$$\lim_{x\searrow a_m} f(x) - \lim_{x\nearrow a_m} f(x) = 2s_m .$$

b) Suppose now that $\delta_n > 0$ for all $n \in \mathbb{N}$ and let $x, y \in \mathbb{R}$ with $x < y$. We have

$$f(y) - f(x) = \sum_{n \geq 1} \delta_n \cdot \overbrace{\left(\operatorname{sign}(y - a_n) - \operatorname{sign}(x - a_n)\right)}^{= 0 \text{ outside of } D,\ = 2 \text{ if } n \in D} = \sum_{n \in D} 2\delta_n \geq 0,$$

where $D = \{ n \in \mathbb{N} : x \leq a_n \leq y \}$. This shows that f is increasing. ∎

c) Note that since g is monotone, at each point $b \in B$ the onesided limits exist, and moreover it holds that

$$\lim_{x \nearrow b} g(x) < \lim_{x \searrow b} g(x).$$

Since $\mathbb{Q}$ is dense in $\mathbb{R}$, for any $b \in B$ we find $q(b) \in \mathbb{Q}$ such that

$$\lim_{x \nearrow b} g(x) < q(b) < \lim_{x \searrow b} g(x).$$

Moreover, for $b_1, b_2 \in B$ with $b_1 < b_2$, we must have $q(b_1) < q(b_2)$ since g is increasing. Thus, $q : B \to \mathbb{Q}$ defines an injective map, which in particular implies that B is (at most) countable. ∎